PROJECT REPORT

METRO TICKET GENERATING SYSTEM IN SERVICENOW

Digital Metro Ticketing with Automated Fare Calculation and QR Codes

Date: 28 September 2026

Student Name: Mohamed Radhuwan

Project Documentation

Technology: ServiceNow
Features: Service Catalog, Flow Designer, Notifications
Project: Metro Ticket Generating System in ServiceNow

TABLE OF CONTENTS

1. INTRODUCTION
2. IDEATION PHASE
3. REQUIREMENT ANALYSIS
4. PROJECT DESIGN
5. PROJECT PLANNING & SCHEDULING
6. FUNCTIONAL AND PERFORMANCE TESTING
7. RESULTS
8. ADVANTAGES & DISADVANTAGES
9. CONCLUSION
10. FUTURE SCOPE
11. APPENDIX

1. INTRODUCTION

1.1 Project Overview

Digitize metro ticket booking using ServiceNow to enable fast, accessible, and automated ticket generation through QR codes. The system allows users to book tickets by selecting stations and travel details, automatically calculates fares, and generates digital tickets instantly. It streamlines metro operations using Service Catalog, Flow Designer, and Notifications for a seamless and efficient travel experience.

The system replaces manual counter-based ticketing with a guided, automated process, reducing queues and errors while creating a foundation for digital payments and data-driven planning.

1.2 Purpose

- Enhance Commuter Convenience: Implement user-friendly digital ticketing options such as mobile apps, QR codes, and WhatsApp-based bookings to reduce wait times and streamline the ticketing process.
- Increase Operational Efficiency: Adopt Automated Fare Collection (AFC) systems to minimize manual interventions, reduce errors, and lower operational costs.
- Promote Digital Payment Adoption: Encourage the use of digital payment methods like UPI, credit/debit cards, and mobile wallets to facilitate cashless transactions and improve revenue collection.
- Facilitate Data-Driven Decision Making: Utilize data analytics from digital transactions to gain insights into passenger behavior, peak usage times, and route optimization opportunities.
- Support Environmental Sustainability: Reduce paper usage by transitioning to digital tickets, contributing to eco-friendly practices and aligning with sustainability goals.

Users: Passengers (end users), Station Managers, Metro Operations Team and IT Admins.

2. IDEATION PHASE

2.1 Problem Statement

Metro passengers face long queues, manual fare errors and paper tickets. Without automation, station and operations teams also lack accurate, digital records. The project solves this with a guided ServiceNow ticket booking process that calculates fares and issues QR code tickets instantly.

2.2 Empathy Map Canvas

Perspective	Project Interpretation
Says	I need a fast way to book a metro ticket.
Thinks	I want the fare to be correct and the QR ticket to scan.
Does	Selects stations, enters travel details and submits.
Feels	Wants a quick and reliable travel experience.
Pain Points	Queues, fare errors, paper tickets.
Needs	Automated fare, instant QR ticket, simple booking.

2.3 Brainstorming

- Create a Service Catalog item for ticket booking with Source, Destination, Passenger Type and Number of Tickets.
- Automate fare calculation with a fare logic script.
- Generate QR code digital tickets.
- Map station details to bookings.
- Use Flow Designer and Notifications for confirmations.
- Build user-facing UI pages.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Passenger Action	System Response	Expected Outcome
Access	Open booking page	Loads catalog form	Passenger can start booking
Select	Choose source, destination, passenger type, quantity	Applies defaults and validation	Valid data captured
Fare	Review fare	Fare Logic Script calculates fare	Correct fare shown
Submit	Submit booking	Ticket record and QR code created	Digital ticket issued
Notify	Receive confirmation	Notification sent	Passenger has ticket
Travel	Show QR at gate	Ticket validated	Passenger travels

3.2 Solution Requirements

Functional Requirements

FR No.	Functional Requirement	Sub-Requirement / Expected Behavior
FR-1	Ticket Booking Catalog Item	Service Catalog item with variables Source, Destination, Passenger Type and Number of Tickets.
FR-2	Field Validation	Non-empty source/destination, fare match and approved travel date.
FR-3	Mandatory Fields	Enforced on client and server side to prevent incomplete submissions.
FR-4	Auto-Populate Logic	Default values (e.g., single or return journey) using catalog client scripts and UI policies.
FR-5	Fare Calculation	Fare is calculated automatically from stations, passenger type and number of tickets.
FR-6	QR Code Generation	A digital ticket with a QR code is generated instantly after booking.
FR-7	Station Details Mapping	Station details are mapped to the booking for fare and validation.
FR-8	Field Mapping	Process Automation Engine auto-maps and populates values entered by the user.
FR-9	Notifications	Booking confirmation sent through Flow Designer and Notifications.
FR-10	UI Pages & Testing	User-facing UI pages, followed by UAT and deployment.

Non-Functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Simple and fast booking experience.
NFR-2	Data Consistency	Validation and mapped fields reduce wrong entries.
NFR-3	Reliability	Booking consistently creates the intended ticket and QR code.
NFR-4	Maintainability	Catalog, scripts, flows and pages are configured separately.
NFR-5	Security	ServiceNow role-based access controls.
NFR-6	Performance	Responsive booking and ticket generation.

3.3 Data Flow Diagram

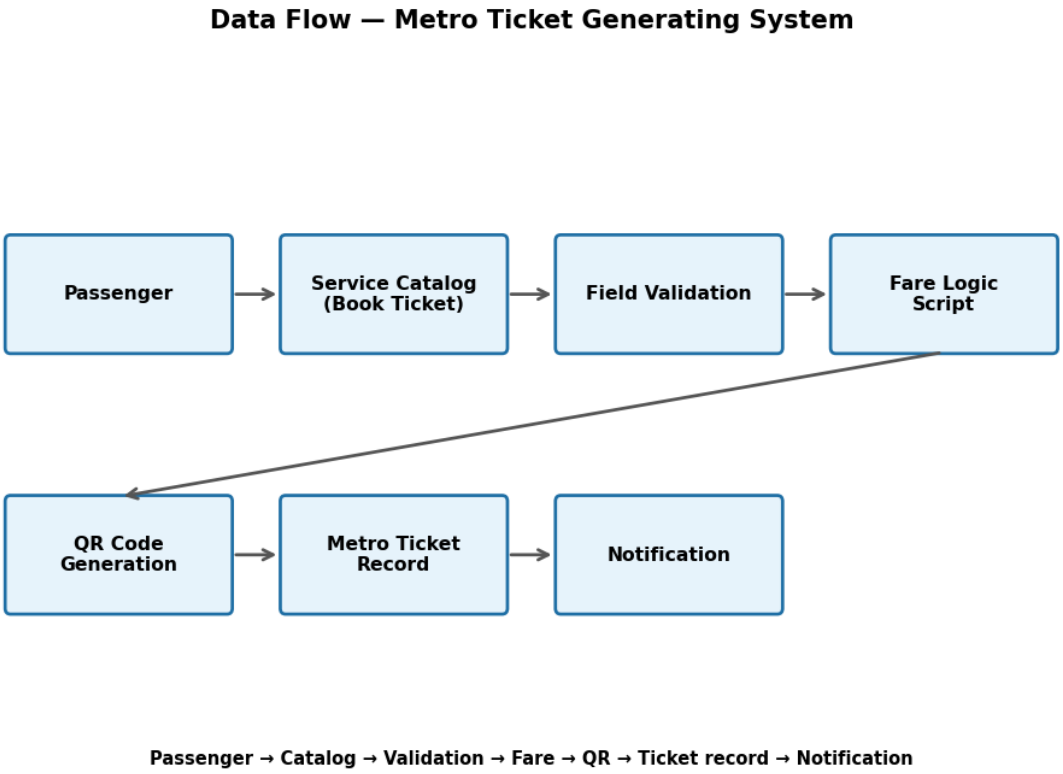


Figure 1: High-level flow from passenger booking to the digital QR ticket.

3.4 Technology Stack

Component	Technology / Configuration	Purpose
Platform	ServiceNow	Application platform.
Catalog	Service Catalog item	Guided ticket booking.
Automation	Flow Designer, Process Automation Engine	Workflow and field mapping.
Scripts	Catalog Client Scripts, Fare Logic Script	Validation, defaults and fare calculation.
Notifications	ServiceNow Notifications	Booking confirmation.
QR Code	QR Code Integration	Digital ticket generation.
Interface	UI Pages	Passenger-facing pages.
Data	Ticket and Station tables	Store bookings and station details.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

Parameter	Project Fit
Problem	Manual ticketing causes queues, errors and paper use.
Solution	Automated ServiceNow ticketing with fare calculation and QR codes.
Customer Benefit	Fast, accessible booking.
Operational Benefit	Fewer manual steps and errors.
Scalability	Payments, analytics and WhatsApp booking can be added.

4.2 Proposed Solution

A Service Catalog item collects booking details; validation, auto-populate and mapping rules keep data consistent; a fare logic script and QR code integration produce the digital ticket; Flow Designer and Notifications inform the passenger.

4.3 Solution Architecture

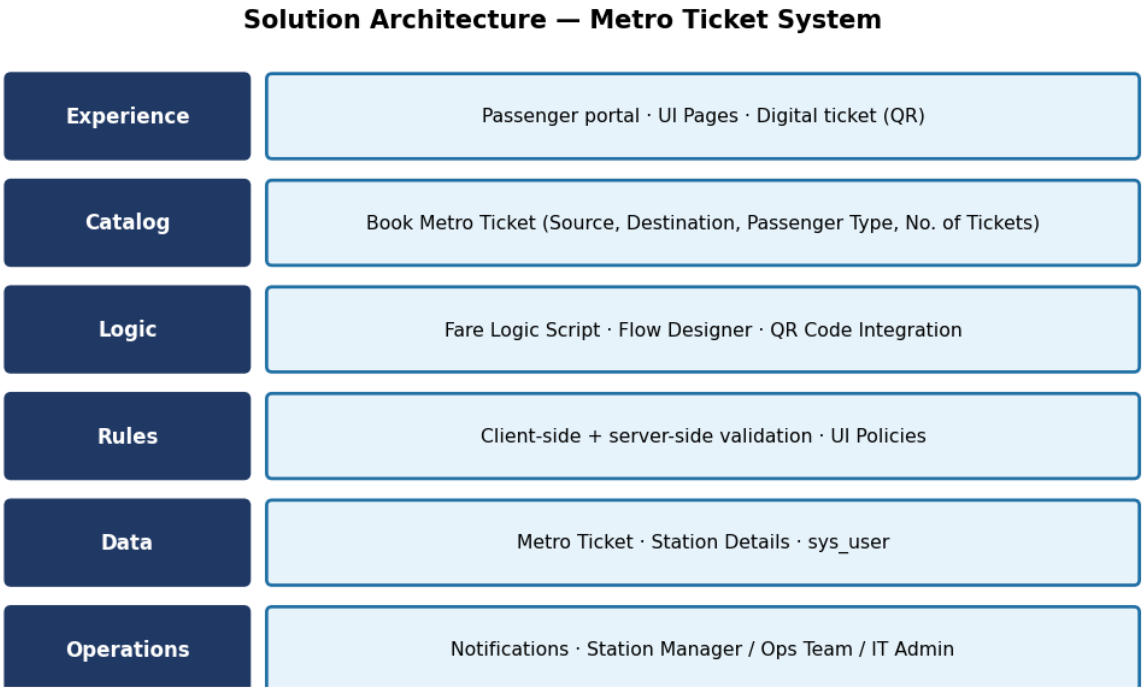


Figure 2: Metro Ticket solution architecture.

Data Architecture

Data architecture is the blueprint for how data is collected, stored, managed, integrated, and used across an organization. It defines the policies, standards, models, and rules that govern how data flows and supports business objectives.

Validation & Data Handling

Item	Description
Field validations	Non-empty source/destination, fare match, approved travel date.
Mandatory fields	Enforced on client and server side to prevent incomplete submissions.
Auto-populate logic	Default values (e.g., single, return journey) using catalog client scripts and UI policies.
Fields mapping	Process Automation Engine auto-maps fields and populates values given by the user/clients.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Execution roadmap milestones: Catalog Form Setup → Fare Logic Script → QR Code Integration → Station Details Mapping → UI Pages → UAT & Deployment.

Phase / Sprint	Major Tasks	Deliverable
Sprint 1 – Catalog Form Setup	Create catalog item and variables.	Booking form.

Phase / Sprint	Major Tasks	Deliverable
Sprint 2 – Fare Logic Script	Fare calculation and validations.	Automatic fare.
Sprint 3 – QR Code Integration	Generate QR code for tickets.	Digital ticket.
Sprint 4 – Station Details Mapping	Map stations and auto-populate.	Mapped station data.
Sprint 5 – UI Pages	Build UI pages, flows and notifications.	Passenger interface.
Sprint 6 – UAT & Deployment	Test and deploy.	Validated solution.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing focused on functional correctness and user interaction rather than load benchmarking.

Test ID	Test Scenario	Expected Result	Status
TC-01	Open booking form	Catalog item loads	Pass
TC-02	Select source and destination	Stations selectable	Pass
TC-03	Empty source/destination	Validation error shown	Pass
TC-04	Default journey type	Single/return default populated	Pass
TC-05	Fare calculation	Fare matches selection	Pass
TC-06	Invalid travel date	Unapproved date rejected	Pass
TC-07	Mandatory fields	Incomplete form blocked	Pass
TC-08	QR code generation	QR code generated on ticket	Pass
TC-09	Notification	Confirmation sent	Pass
TC-10	End-to-end	Complete booking flow works	Pass

Performance note: formal concurrent-user/load testing was not part of the supplied implementation evidence.

7. RESULTS

7.1 Output Screenshots

Screenshots are reserved for the final submission. Insert the following:

Figure	Recommended Screenshot
Figure 3	Ticket booking catalog form.
Figure 4	Source, destination and passenger type selection.
Figure 5	Validation messages for missing fields.
Figure 6	Automatically calculated fare.
Figure 7	Generated digital ticket with QR code.
Figure 8	Booking confirmation notification.
Figure 9	Metro ticket record in ServiceNow.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Fast and accessible ticket booking.
- Automated fare calculation reduces errors.
- QR code digital tickets reduce paper use.

- Supports digital payments and data-driven decisions.

Disadvantages

- Depends on ServiceNow instance availability and internet access.
- Payment gateway and WhatsApp booking are not yet implemented.
- No formal load testing performed.

9. CONCLUSION

The Metro Ticket Generating System in ServiceNow digitizes ticket booking end to end, from a validated catalog form to automatic fare calculation and an instant QR code ticket. It improves convenience for passengers and efficiency for metro operations.

10. FUTURE SCOPE

- Integrate UPI, card and wallet payments.
- Mobile app and WhatsApp-based booking.
- Analytics dashboards for peak usage and route optimization.
- Automated gate validation of QR tickets.

11. APPENDIX

Project: Metro Ticket Generating System in ServiceNow. Roles: Passengers, Station Managers, Metro Operations Team, IT Admins. Technology: ServiceNow.